

**RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL**

**New Scheme Based On AICTE Flexible Curricula**

**Artificial Intelligence and Data Science, V-Semester**

**AD-501 Theory of Computation**

**COURSE OBJECTIVES:**

This course will help students to learn several formal mathematical models of computation along with their relationships with formal languages and grammars. Students will also learn about solvable and unsolvable problems.

**COURSE OUTCOMES:**

**After completing the course student should be able to:**

1. Compare and analyze different theoretical computational models, languages and grammars.
2. Design and construct finite automata, pushdown automata and Turing machine for various problems.
3. Identify limitations of some computational models and possible methods of proving them.
4. Describe the concept of computable and non-computable problems.

**Unit-I**

**Introduction of Automata Theory:** Review of Sets, Mathematical formal proofs including proof by induction and by contradiction, Introduction to languages, grammars and automata: Alphabet, Representation of language and grammar, Types of Automata, Finite Automata as a language acceptor and translator, Moore machines and mealy machines, composite machine, Conversion from Mealy to Moore and vice versa.

**Unit-II**

**Types of Finite Automata:** Non Deterministic Finite Automata (NFA), Deterministic finite automata machines, conversion of NFA to DFA, minimization of automata machines, regular expression, applications of regular expressions, Arden's theorem. Meaning of union, intersection, concatenation and closure, 2 way DFA.

**Unit-III**

**Grammars:** Types of grammar, context sensitive grammar, and context free grammar, regular grammar. Derivation trees, ambiguity in grammar, simplification of context free grammar, conversion of grammar to automata machine and vice versa, Chomsky hierarchy of grammar, Chomsky normal form and Greibach normal form.

## Unit-IV

**Push down Automata:** example of PDA, deterministic and non-deterministic PDAs, Context Free Grammar, Parsing, Ambiguity, Normal form of CFGs, CFG to NPDA, NPDA to CFGs, CFG equivalent to PDA, Petri nets model.

## Unit-V

**Turing Machine:** Turing Machine as acceptor, Recognizing a Language, Universal TMs, Linear Bounded Automata, Context Sensitive Languages, Recursive and Recursively Enumerable Languages, Unrestricted Grammars. Halting problem of Turing machine & the post correspondence problem, Concept of Solvability and Unsolvability, Church's Thesis, Complexity Theory – P and NP problems.

## RECOMMENDED BOOKS

1. Hopcroft, Ullman, Motwani, "Introduction to Languages, Automata and Computation", 3<sup>rd</sup> Edition, Pearson Education, 2008.
2. John C. Martin, "Introduction to Languages and the Theory of Computation", Fourth Edition, Mc Graw Hill, 2010.
3. Peter Linz, "An Introduction to Formal Languages and Automata", Sixth Edition, Jones and Bartlett, 2016.
4. Lewis and Papadimitriou, "Elements of Theory of Computation", Second Edition, Pearson Education, 2015.
5. K.L.P. Mishra and N. Chandrasekaran, "Theory of Computer Science: Automata, Languages and Computation", Third Edition, Prentice Hall, 2006.
6. Cohen John, "Introduction to Computer Theory", Second Edition, Wiley and Sons, 2007.
7. Theory of Computation, Wood, Harper & Row.

## LIST OF EXPERIMENTS

1. Design a Program for creating machine that accepts three consecutive one.
2. Design a Program for creating machine that accepts the string always ending with 101.
3. Design a Program for Mode 3 Machine
4. Design a program for accepting decimal number divisible by 2.
5. Design a program for creating a machine which accepts string having equal no. of 1's and 0's.
6. Design a program for creating a machine which count number of 1's and 0's in a given string.
7. Design a Program to find 2's complement of a given binary number.
8. Design a Program which will increment the given binary number by 1.
9. Design a Program to convert NDFA to DFA.
10. Design a Program to create PDA machine that accept the well-formed parenthesis.
11. Design a PDA to accept WCWR where w is any string and WR is reverse of that string and C is a Special symbol.
12. Design a Turing machine that's accepts the following language  $a^n b^n c^n$  where  $n > 0$ .